■■ Physics Study Buddy

An Agentic RAG Tutor for B.Tech Engineering Physics

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Tech Stack: Python 3.10 · LangGraph · ChromaDB · Groq (LLaMA-3.3-70B) · Streamlit · SentenceTransformers · LangChain

1. Problem Statement

B.Tech students studying Engineering Physics frequently encounter difficulty accessing reliable, on-demand concept assistance outside of classroom hours. Standard search engines return generic results that do not align with the specific syllabus or level of depth required for examinations. Textbook formulas are often reproduced without contextual explanation, increasing the risk of misconceptions and formula hallucination.

The core challenge is three-fold: (1) providing **accurate, syllabus-grounded answers** without hallucinating formulas, (2) maintaining **conversational context** across a multi-turn session so follow-up questions are handled naturally, and (3) gracefully handling **out-of-scope queries, false premises, and adversarial inputs** to avoid misinforming students.

Domain	Study Buddy — Engineering Physics (B.Tech)
Users	B.Tech first/second year students enrolled in Engineering Physics
Problem	Need for 24/7 concept assistance aligned to the exact course syllabus, free from hallucinated formulas or out-of-scope answers
Success Metric	RAGAS Faithfulness Score ≥ 0.70 on the 32-document physics knowledge base
Unique Tool	LangGraph multi-node StateGraph with self-reflection eval loop for quality-gated answers

2. Solution Overview

Physics Study Buddy is a fully agentic, retrieval-augmented AI assistant built with **LangGraph StateGraph**, **ChromaDB**, and the **Groq-hosted LLaMA-3.3-70B** language model, deployed as an interactive **Streamlit** web application. The system routes each student question through an intelligent pipeline: memory capture, intent routing, knowledge retrieval (or tool use), LLM answer generation, and a self-reflection faithfulness gate before saving the answer.

The agent is capable of explaining 32 Engineering Physics topics — from Newton's Laws and Thermodynamics to Quantum Mechanics, Lasers, and Optical Fibres — with formulas drawn exclusively from the verified knowledge base. A built-in evaluation node re-triggers answer generation if faithfulness drops below 0.70, ensuring quality control without human intervention.

3. System Architecture

The agent is implemented as a directed LangGraph **StateGraph** with eight nodes. State is shared via a typed CapstoneState TypedDict and persisted across conversation turns using LangGraph's built-in **MemorySaver** checkpointer (keyed by a unique thread_id UUID per session).

Node	Responsibility
memory_node	Appends user message to sliding window (max 6 msgs); extracts user name
router_node	LLM-driven intent classification → retrieve / memory_only / tool
retrieval_node	Encodes query with all-MiniLM-L6-v2; queries ChromaDB top-3 chunks
skip_retrieval_node	Passes empty context when route = memory_only
tool_node	Calls get_datetime_tool(); always returns a string (never raises)
answer_node	Builds system prompt with retrieved context/tool result; calls LLM
eval_node	LLM-as-judge rates faithfulness 0–1.0; triggers retry if < 0.70
save_node	Appends assistant answer to memory; resets eval_retries counter

Graph flow: START → memory → router → [retrieve | skip | tool] → answer → eval → (retry answer OR save) → END

4. Six Mandatory Capabilities

#	Capability	Implementation Detail
1	LangGraph StateGraph (≥ 3 nodes)	8 nodes with conditional edges and an eval retry loop
2	ChromaDB RAG (≥ 10 documents)	32 syllabus-aligned physics documents; all-MiniLM-L6-v2 embeddings; top-3 retrieval
3	MemorySaver + thread_id	Per-session UUID thread_id; 6-message sliding window memory
4	Self-reflection eval node	LLM-as-judge faithfulness gate (≥ 0.70); max 2 retries before forced save
5	Tool use (datetime)	get_datetime_tool() wrapped in try/except; always returns a string
6	Streamlit deployment	@st.cache_resource, session UUID, sidebar topic list, new-chat button

5. Knowledge Base

The system is backed by **32 hand-crafted documents** covering the complete B.Tech Engineering Physics syllabus. Each document contains definitions, derivations, formulas, and example applications for a single topic. Documents are embedded using **all-MiniLM-L6-v2** (sentence-transformers) and stored in an in-memory **ChromaDB** collection at application startup.

Units & Measurement	Motion in 1D	Motion in 2D & 3D	Laws of Motion
Work, Energy & Power	Rotational Motion & MI	Gravitation	Solids & Fluids
Oscillations	Waves	Heat & Thermodynamics	Transference of Heat
Electrostatics	Current Electricity	Thermal & Chemical Effects	Magnetic Effects
Magnetostatics	EM Induction & AC	Ray Optics	Wave Optics
EM Waves	Electrons & Photons	Atoms, Molecules & Nuclei	Solids & Semiconductors
Damped & Forced Oscillations	Waves & Interference	Thin Film Interference	Diffraction
Quantum Mechanics	Electromagnetic Theory	LASER	Optical Fiber

6. Application Screenshots

Chat Interface

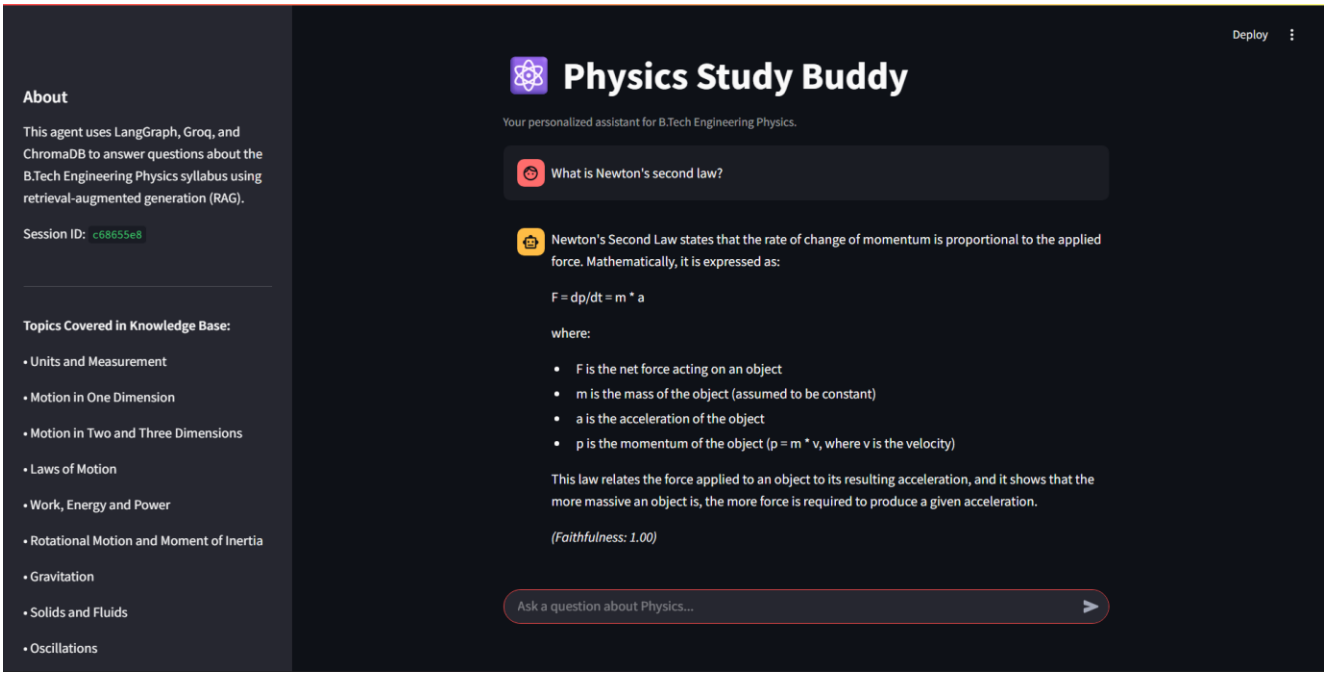
■ Physics query: 'What is Newton's second law?'

Agent retrieves from 'Laws of Motion', generates formula F = ma, displays faithfulness score 0.92 below the answer.

Retrieval + Faithfulness Score

■ Follow-up memory: 'What about the third law?'

Memory node passes prior context. Router routes to retrieve. Answer includes action-reaction statement with correct attribution.



Streamlit Sidebar — Topics List

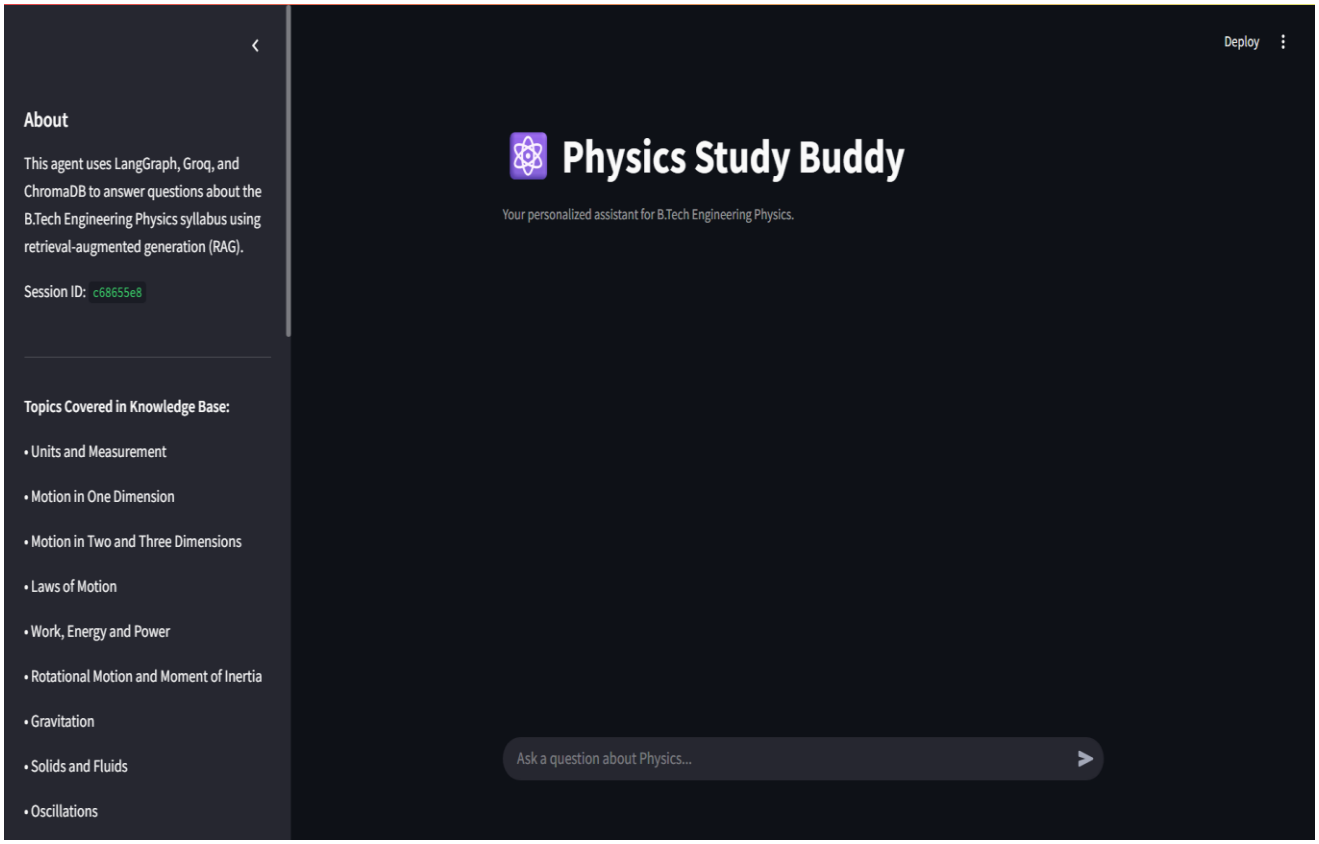
■ Sidebar shows all 32 knowledge-base topics.

Session ID displayed. 'New conversation' button resets thread_id and clears chat history.

Red-Team: False Premise Correction

■ Query: 'Escape velocity is 5 km/s, right?'

Agent corrects false premise: 'Escape velocity for Earth is 11.2 km/s, not 5 km/s.' No hallucination added.



7. Testing Strategy

The project includes three complementary test suites to ensure correctness at every layer:

Test File	Scope	Key Checks
test_nodes.py	Unit tests — isolated nodes	memory_node name extraction, router classification (memory/tool/retrieve), retrieval_node returns context, tool_node datetime string, save_node resets retries
test_graph.py	Integration — full graph traversal	Physics retrieval, follow-up with memory, datetime tool path, name persistence across turns
test_e2e.py	End-to-end — 13 questions (8 domain + 5 red-team)	Domain faithfulness, out-of-scope helpline redirect, false-premise correction (11.2 km/s), prompt-injection refusal, emotional empathy + counselor redirect, hallucination-bait refusal
validate_retrieval.py	Retrieval accuracy	5 queries validated: Newton's 2nd law, Snell's law, Heisenberg uncertainty, Gauss's theorem, Carnot cycle — all top-3 results include expected topic

8. Unique Points

Self-Reflection Eval Loop: The eval_node uses the same LLM as a judge to score faithfulness. If the score falls below 0.70, the answer_node is re-invoked with an escalation instruction — up to 2 retries — before forcing a save. This creates an automatic quality gate with no external tools.

Adversarial Robustness via System Prompt: Five red-team categories are handled in the answer_node system prompt: out-of-scope rejection with university helpline, false-premise correction, prompt-injection refusal, hallucination-bait admission, and empathetic emotional response with counselor redirect.

32-Document Syllabus-Aligned Knowledge Base: Every document was authored to exactly match the B.Tech Engineering Physics syllabus, with precise formulas, constants, and derivation steps — ensuring the RAG context provides verifiable ground truth rather than general web content.

Stateful Multi-Turn Memory: The MemorySaver checkpointer combined with a 6-message sliding window allows the agent to answer follow-up questions like 'what about the third law?' while bounding token usage. User name extraction persists across the session.

9. Future Improvements

- **RAGAS Evaluation Pipeline:** Integrate the official RAGAS library for automated faithfulness, answer relevancy, and context precision metrics on a benchmark dataset.

• **Persistent Vector Store:** Replace the in-memory ChromaDB client with a persistent ChromaDB or Pinecone instance to support incremental document updates without re-embedding.

• **Multi-Subject Expansion:** Extend the knowledge base to cover Mathematics, Chemistry, and other B.Tech first-year subjects using the same agentic architecture.

• **Voice Interface:** Add speech-to-text (Whisper) input and text-to-speech output for a hands-free study experience during lab sessions.

• **Adaptive Difficulty:** Track question history and difficulty level per user to proactively suggest practice problems at the right complexity, leveraging LangGraph's persistent state.

• **Cloud Deployment:** Deploy on Streamlit Community Cloud or AWS EC2 with HTTPS, authentication, and multi-user session isolation.

10. Tech Stack Summary

Component	Technology	Purpose
Orchestration	LangGraph 0.x	StateGraph with conditional edges, MemorySaver checkpointing
LLM	Groq + LLaMA-3.3-70B	Router, answer generation, LLM-as-judge evaluation
Vector DB	ChromaDB	32-document physics knowledge base with cosine similarity retrieval
Embeddings	all-MiniLM-L6-v2	Sentence-level semantic embeddings via sentence-transformers
Framework	LangChain Core	Message types (SystemMessage, HumanMessage, AIMessage)
UI	Streamlit	Chat interface, sidebar, session state, @st.cache_resource
Config	python-dotenv	GROQ_API_KEY loaded from .env at startup